
Additive Relational Bindings in Transformers: What Sparse Autoencoders Miss

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Abstract

Language models often need to represent which entities are bound to which attributes, as in “Alice lives in Paris. Bob lives in London.” How models construct such binding representations is poorly understood, and it remains unclear whether sparse autoencoders (SAEs) recover the binding representations that models actually use. We train a 2-layer attention-only transformer on a synthetic relational retrieval task and reverse-engineer the circuit that solves the task perfectly. We find that Layer 0 writes an approximately additive entity–relation address and a separate payload at each fact slot, while Layer 1 retrieves the matching payload by same-head query-key matching against these addresses. Linear probes decode the joint address with 100% accuracy, an additive decomposition explains 99.8% of its variance, and causal patches over the address flip predictions to a distractor. However, SAEs trained on the same activation site do not recover the joint address as clean individual features, despite reconstructions preserving full task accuracy. This provides a concrete example of a composed representation that is linearly decodable and causally used, yet not cleanly exposed as sparse features by an SAE.

1. Introduction

Language models often face tasks where correctness depends on preserving bindings between variables, not merely representing the variables themselves. In a context such as “Alice lives in Paris. Bob lives in London,” the model must keep track of which city belongs to which person. This is an instance of relational composition: multiple pieces of information must be combined into a structured representation that can later be queried. Wattenberg & Viégas (2024) argue that such composition mechanisms are still poorly un-

derstood and that different forms of relational composition have different implications for feature-based interpretability methods such as Sparse Autoencoders (SAEs).

Prior work provides evidence that language models maintain entity bindings in context. Feng & Steinhardt (2023) identify binding-ID-like directions used to associate entities with attributes across several model families, and Prakash et al. (2024) trace entity-tracking mechanisms in pretrained and fine-tuned models. Recent work on multi-entity “slots” further shows that models can represent multiple entities at a single token position, while emphasizing a distinction between information that is linearly available and information the model functionally uses (Bogdan & Lindsey, 2026). These results motivate a narrower mechanistic question: when a model uses an address-like representation to retrieve a bound value, how is that address constructed, how is it consumed by the circuit, and can it be recovered by sparse feature methods?

We study this question in a controlled toy transformer trained on a synthetic relational retrieval task. Controlled models are useful when the goal is to identify a mechanism rather than maximize scale. The results of toy models have previously been informative for work on larger models, like on modular arithmetic (Nanda et al., 2023; Baeumel et al., 2025) and superposition (Elhage et al., 2022; Bricken et al., 2023). More closely related, recent toy-model work on relative-magnitude relational composition shows that attention-only transformers can learn nontrivial composition mechanisms with implications for the common view of SAE latents as binary, independent features (Farrell et al., 2025). Our setting targets a different mechanism. Instead of ordered list copying, the model must retrieve a payload entity by constructing and matching an entity–relation address.

We make three contributions. First, we train a 2-layer attention-only transformer on a synthetic relational retrieval task and identify the circuit that solves it. Second, we show that one Layer 0 attention head represents each fact address as the sum of an entity component and a relation component, while Layer 1 uses this address to match the query to the correct fact slot. Third, we find a gap between linear decodability and SAE feature recovery: the composed address is perfectly linearly decodable and causally controls

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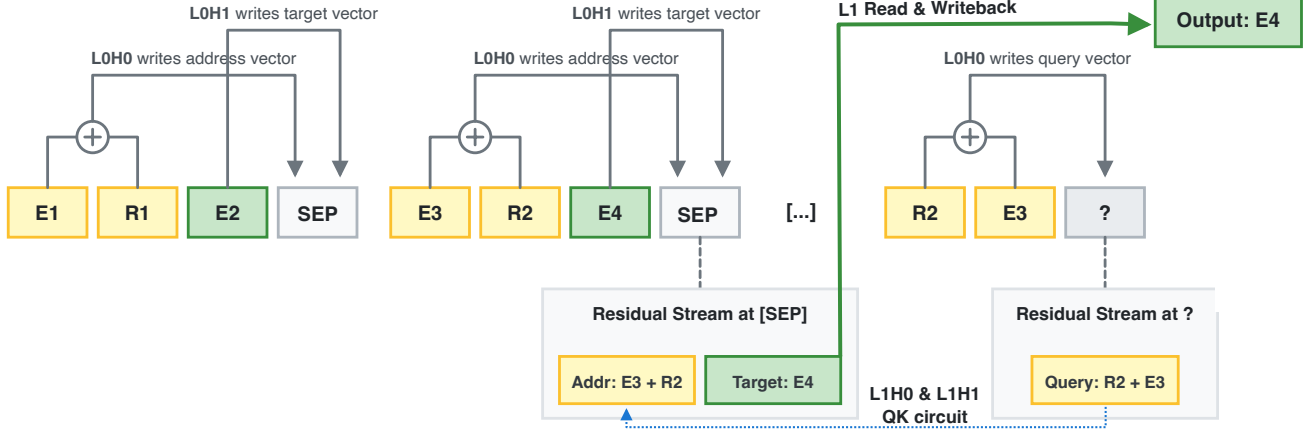


Figure 1. Staged retrieval circuit. At each fact, the first head of Layer 0 (LOH0) writes an address representation (E_1, R), the second (LOH1) a payload representation E_2 to the $[SEP]$ position. At the query, LOH0 writes the queried address (E_q, R_q) to the $[?]$ position. The two Layer 1 heads (L1H0 and L1H1) then match the query against fact addresses via their QK circuits and retrieve the corresponding payload.

retrieval, but is not cleanly recovered as individual SAE features. Dataset, code, and model checkpoints are available at: <https://anonymous.4open.science/r/relational-composition-ssk>.

2. Methods

2.1. Model

We train a 2-layer, 2-head attention-only transformer with residual-stream dimension $d_{\text{model}} = 256$ and head dimension $d_{\text{head}} = 128$. The model uses Rotary Positional Embeddings (RoPE) and layer normalization. To force targeted entity retrieval, the output vocabulary is restricted to the 100 entity tokens.

2.2. Task

We study a synthetic relational retrieval task designed to isolate entity–relation binding. Let $\mathcal{E}$ be a set of 100 entity tokens and $\mathcal{R}$ a set of 10 relation tokens. We additionally use two dedicated synthetic tokens: a separator token $[SEP]$ and a query marker $[?]$. Each prompt contains $n \in \{4, \dots, 8\}$ facts, serialized as

$$E_{1,i} R_i E_{2,i} [SEP],$$

where $E_{1,i}, E_{2,i} \in \mathcal{E}$ and $R_i \in \mathcal{R}$. The prompt ends with a query

$$R_q E_q [?],$$

and the target is the entity $E_{2,j}$ from the unique fact whose address matches the query, i.e. $(E_{1,j}, R_j) = (E_q, R_q)$. The model is trained only on the final answer position, with output logits restricted to the 100 entity tokens.

The task is constructed to require matching on the composed address (E_1, R) , rather than on either variable alone. Within

each prompt, $(E_{1,i}, R_i)$ pairs are unique. In addition, with probability 0.75, we include a distractor fact, sharing the query entity but with a different relation and target. Thus, solving the task requires locating the fact slot whose entity–relation address matches the query and returning its payload entity E_2 .

The reversed query order $R_q, E_q, [?]$, rather than $E_q, R_q, [?]$, is a small but deliberate design choice. If the query repeated the fact prefix order, the task would more closely resemble an induction-head-style continuation problem: after seeing $E_1 R E_2$ in context, the model could answer $E_1 R [?]$ by matching a repeated local token pattern. By querying in the order $R_q, E_q, [?]$, we make the query differ from the surface order of the facts, reducing this shortcut and forcing the model to match the composed entity–relation address more explicitly.

3. Results

The trained model solves the task perfectly, reaching 100% accuracy on the 20,000-example validation set used during training and 100% accuracy on a separate held-out test set of 8,000 prompts. Unless otherwise stated, all analyses in this section are computed on this held-out test set.

We find that the model solves the task with a staged retrieval circuit (Figure 1). Layer 0 writes two distinct representations to each $[SEP]$ position: an address representation for the fact identity (E_1, R) , and a payload representation for the target entity E_2 . On the query side, Layer 0 also writes the queried address (E_q, R_q) to the final $[?]$ position. Layer 1 then compares the query representation at $[?]$ to the address representations at previous $[SEP]$ positions, selects the matching fact slot, and retrieves the corresponding payload.

Target	Position	Activation site	Accuracy
<i>Fact-side probes after Layer 0 at [SEP] positions</i>			
E_2	[SEP]	Residual stream after Layer 0	100.0%
E_2	[SEP]	Residual stream before Layer 0	1.13%
E_2	[SEP]	Layer 0 head 0 output	0.0%
E_2	[SEP]	Layer 0 head 1 output	100.0%
E_1	[SEP]	Layer 0 head 0 output	100.0%
R	[SEP]	Layer 0 head 0 output	100.0%
(E_1, R)	[SEP]	Residual stream after Layer 0	100.0%
(E_1, R)	[SEP]	Layer 0 head 0 output	100.0%
<i>Query-side probes after Layer 0 at the [?] position</i>			
E_q	[?]	Layer 0 head 0 output	100.0%
R_q	[?]	Layer 0 head 0 output	100.0%

Table 1. Linear probe results. The first head of Layer 0 carries address-side information, while the second carries payload information.

3.1. Layer 0 separates address and payload

Linear probes localize the variables used by the circuit (Table 1). At [SEP] positions after Layer 0, E_2 is perfectly decodable from the residual stream, but not from the residual stream before Layer 0. Head-level probes separate the two roles: Layer 0 head 0 contains the address-side variables E_1 , R , and the joint address (E_1, R) , while Layer 0 head 1 contains the payload E_2 . The query-side variables E_q and R_q are also decodable from the Layer 0 head-0 output at the final [?] position.

3.2. The address decomposes additively

Investigating how the address (E_1, R) is represented, we find that, at the level of (E_1, R) class means, the address is almost exactly explained by an additive decomposition into entity and relation components.

Let $z_i \in \mathbb{R}^{d_{\text{head}}}$ denote the output of Layer 0 head 0 at a [SEP] position for a fact with head entity $E_1 = e$ and relation $R = r$. For each of the 1,000 entity–relation pairs, we compute the class-mean address vector

$$a_{e,r} = \mathbb{E}[z_i \mid E_1 = e, R = r].$$

We then fit the additive model

$$\hat{a}_{e,r} = \mu + u_e + v_r,$$

where μ is a global offset, u_e is an entity component, and v_r is a relation component. This model explains 99.8% of the pair-level variance:

$$\text{FVE} = 1 - \frac{\sum_{e,r} \|a_{e,r} - \hat{a}_{e,r}\|_2^2}{\sum_{e,r} \|a_{e,r} - \bar{a}\|_2^2} = 0.998.$$

The mean cosine similarity between actual and predicted class-mean address vectors is 0.9996. To test that this is genuine factorization rather than a post-hoc fit, we held out 200

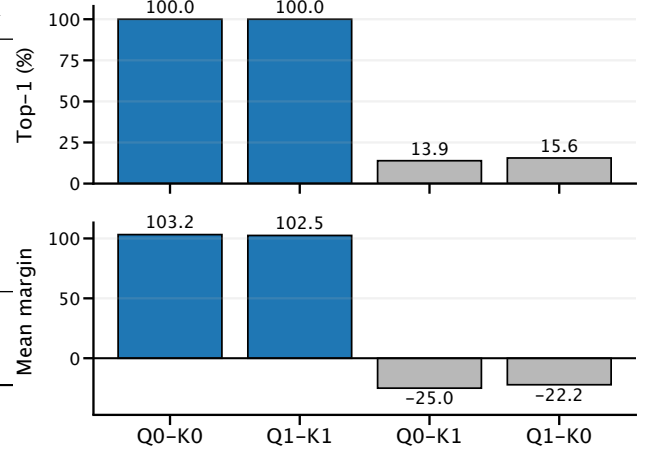


Figure 2. Layer 1 query-key matching. Same-head comparisons select the correct separator with 100% top-1 accuracy and large positive margins, while cross-head comparisons fail. Here $Q0-K0$ denotes scores computed from the query and key vectors of Layer 1 head 0, and $Q1-K1$ for Layer 1 head 1.

of the 1,000 (E_1, R) pairs and fit $\mu + u_e + v_r$ on the remaining 800. The additive model generalized almost perfectly to held-out pairs, achieving FVE 0.998 and mean cosine similarity 0.9995. Applying the same class-level prediction to individual activations still explains 97.7% of activation-level variance, with mean cosine similarity 0.9934.

3.3. Layer 1 retrieves by same-head query-key matching

We next test whether Layer 1 uses the Layer 0 address representations for retrieval. For each Layer 1 head h , we compute the query-key score between the query vector at the final [?] position and the key vector at each preceding [SEP] position:

$$s_h(p) = q_{[?]}^{(h)} \cdot k_p^{(h)}.$$

Here p ranges over separator positions, and p^* denotes the separator belonging to the correct fact. We define the retrieval margin for head h as

$$m_h = s_h(p^*) - \max_{p \neq p^*} s_h(p).$$

This same-head query-key score selects the correct [SEP] position in 100% of the 8,000 held-out prompts for both Layer 1 heads (Figure 2). Head 0 has a mean retrieval margin of +103.2, and head 1 has a mean retrieval margin of +102.5. As a control, we also compare query vectors from one head with key vectors from the other head; these cross-head scores fail, with top-1 rates near chance and negative margins.

3.4. Address interventions redirect retrieval

Finally, we test whether the address representation causally controls which payload is retrieved (Figure 3). We construct

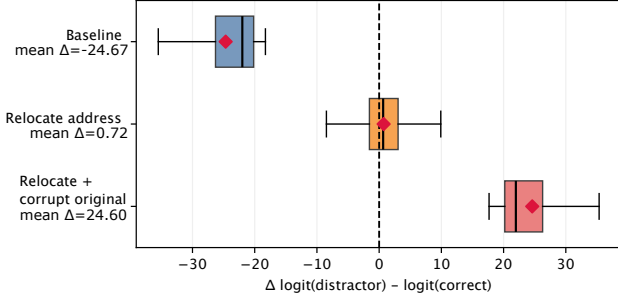


Figure 3. Causal interventions on the address representation. Relocating the target-compatible address to a distractor separator makes the model uncertain; relocating it while corrupting the original target address flips the prediction to the distractor.

mean Layer 0 head-0 address vectors and patch a target-compatible address onto a distractor [SEP] position. This intervention alone makes the model uncertain between the original target and the distractor. When we additionally corrupt the original target address, the model decisively flips to the distractor payload. Defining the logit difference as

$$\Delta = \ell_{\text{distractor}} - \ell_{\text{target}},$$

the mean logit difference moves from strongly favoring the original target ($\Delta = -24.46$) to strongly favoring the distractor ($\Delta = +24.49$). The near-symmetric flip is consistent with the address acting as an additive, location-independent key.

3.5. SAE features do not cleanly recover the joint address

The circuit analysis identifies the Layer 0 head-0 output at [SEP] positions as the most localized activation site for the relational address, and linear probes decode the joint (E_1, R) address perfectly from this activation (see Table 1). We train BatchTopK SAEs (Bussmann et al., 2024) with 1,024 or 2,048 latents and $k = 8$ on two Layer 0 activation sites at [SEP] positions: the address-carrying head-0 output and the residual stream after Layer 0. Replacing these activations with SAE reconstructions preserves 100% test accuracy across all SAE widths and activation sites. We therefore ask whether the relational address is recovered as individual SAE features.

For each of the 1,000 (E_1, R) labels, we select the SAE feature with the highest F1 score for detecting that label from feature activation. Figure 4 shows the 1,024-latent SAE trained on the address-carrying head output, where the contrast with the probe result is sharpest: the best-matching SAE features for the joint address have high recall but low precision, firing on many occurrences of a given address but also on many other addresses. The mean best-feature F1 for joint addresses is only 0.090, despite perfect linear

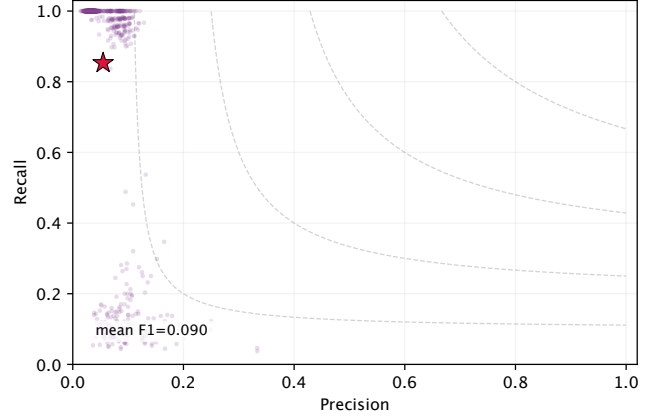


Figure 4. SAE feature recovery of the joint address (E_1, R) from Layer 0 head-0 output at [SEP] positions. Each point shows the precision and recall of the best-F1 SAE feature for one entity-relation pair; the red star marks the per-panel mean.

decodability from the same activation. Appendix Figures 5–8 show that this conclusion is not resolved by doubling SAE width. The residual-stream SAEs recover the payload E_2 much more cleanly and recover the joint address somewhat better than head-output SAEs, but the joint address remains poorly recovered as individual features compared to the linear probe and causal-intervention results.

4. Discussion

In a controlled relational retrieval setting, we find that a small attention-only transformer learns an explicit staged mechanism: Layer 0 constructs entity-relation addresses and payload representations, while Layer 1 retrieves by matching the query address against stored fact addresses. The learned address is almost exactly additive and decomposes into an entity component plus a relation component. Together, these results give a concrete example of relational composition that is linearly decodable, causally used, and can be reverse-engineered at the circuit level. The SAE results show a gap between functional reconstruction and feature-level recovery. Although SAE reconstructions preserve perfect task accuracy, individual SAE features do not cleanly recover the joint address that is linearly decodable and causally controls retrieval. The main limitation is that our evidence comes from a deliberately simplified synthetic setting. This makes the learned mechanism unusually tractable, but does not show whether pretrained language models use analogous address-payload circuits. Future work should test for related mechanisms in fine-tuned or pretrained LLMs, and examine whether the gap between causal use and SAE feature recovery persists at scale.

Impact Statement

This paper gives a circuit-level account of how a small transformer implements relational binding. The methodological point generalizes beyond this setting where a representation can be linearly decodable while not appearing as individual SAE features, because additive composition and the sparse-feature ontology pick out different kinds of structure. Our evidence comes from a synthetic task and a 2-layer model; whether pretrained language models implement entity binding via analogous additive addresses is an open empirical question, and a prerequisite for safety-relevant conclusions from this line of work.

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A. Appendix

SAE width 1,024.

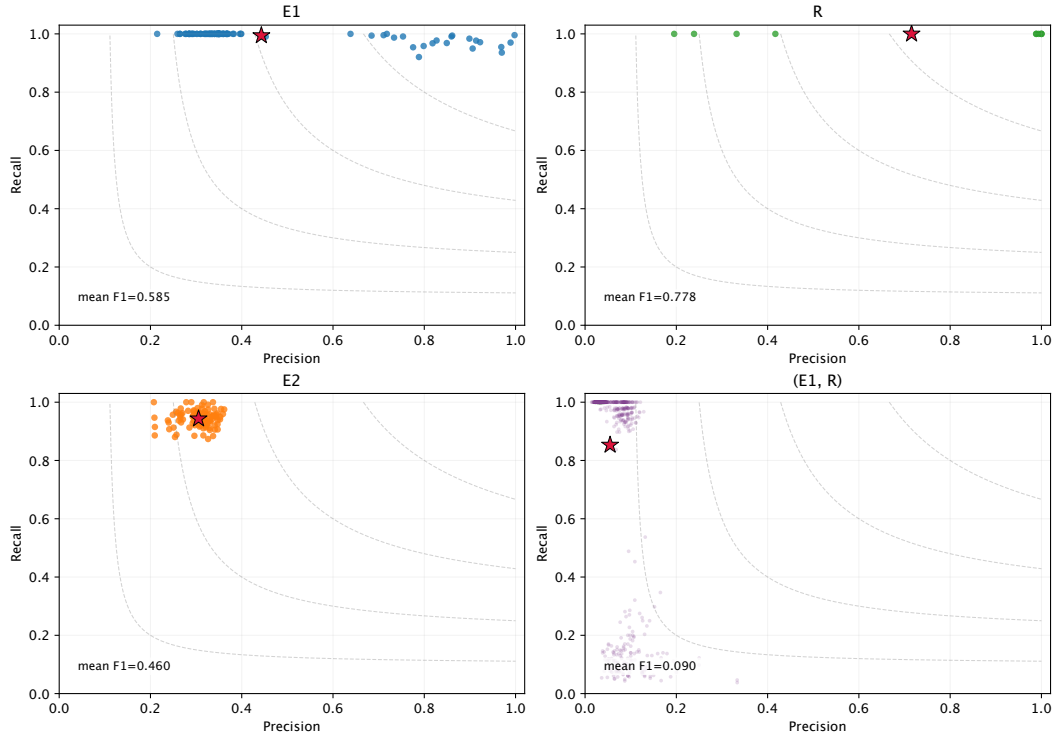


Figure 5. Best-F1 feature recovery for a 1,024-latent SAE trained on the address-carrying Layer 0 head-0 output at [SEP] positions. Panels show results for the marginal labels E_1 , R , E_2 , and the joint address (E_1, R) . Each point is the precision and recall of the highest-F1 SAE feature for one label; the red star marks the per-panel mean.

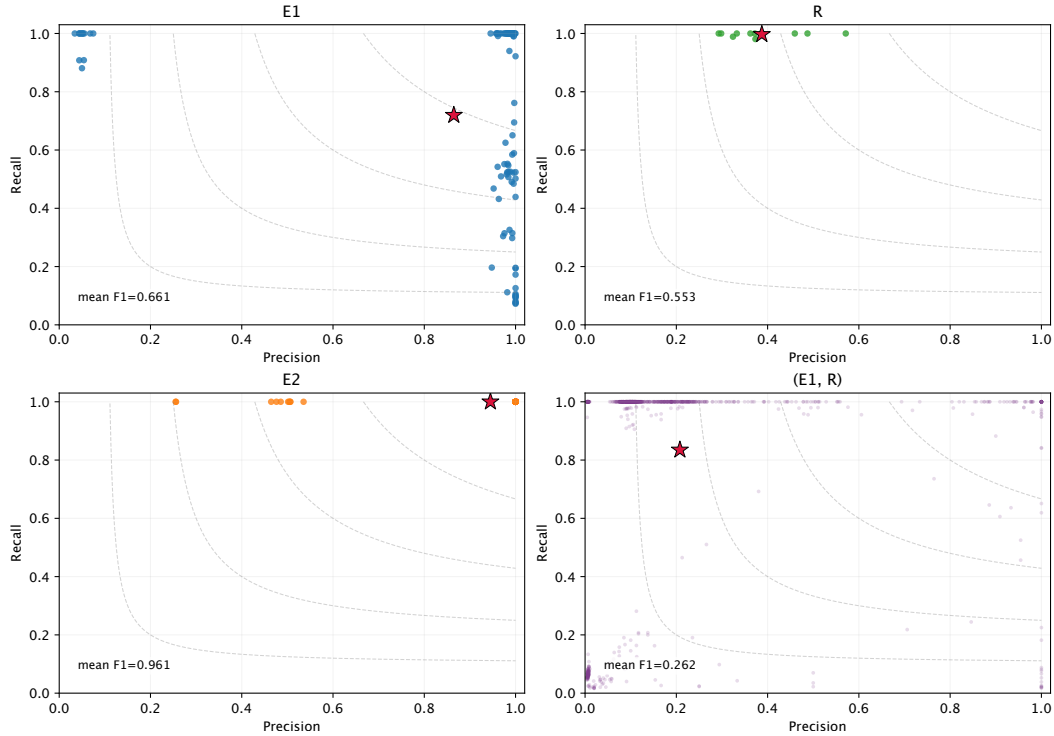


Figure 6. Best-F1 feature recovery for a 1,024-latent SAE trained on the Layer 0 residual stream at [SEP] positions. Panels show results for the marginal labels E_1 , R , E_2 , and the joint address (E_1, R) .

SAE width 2,048.

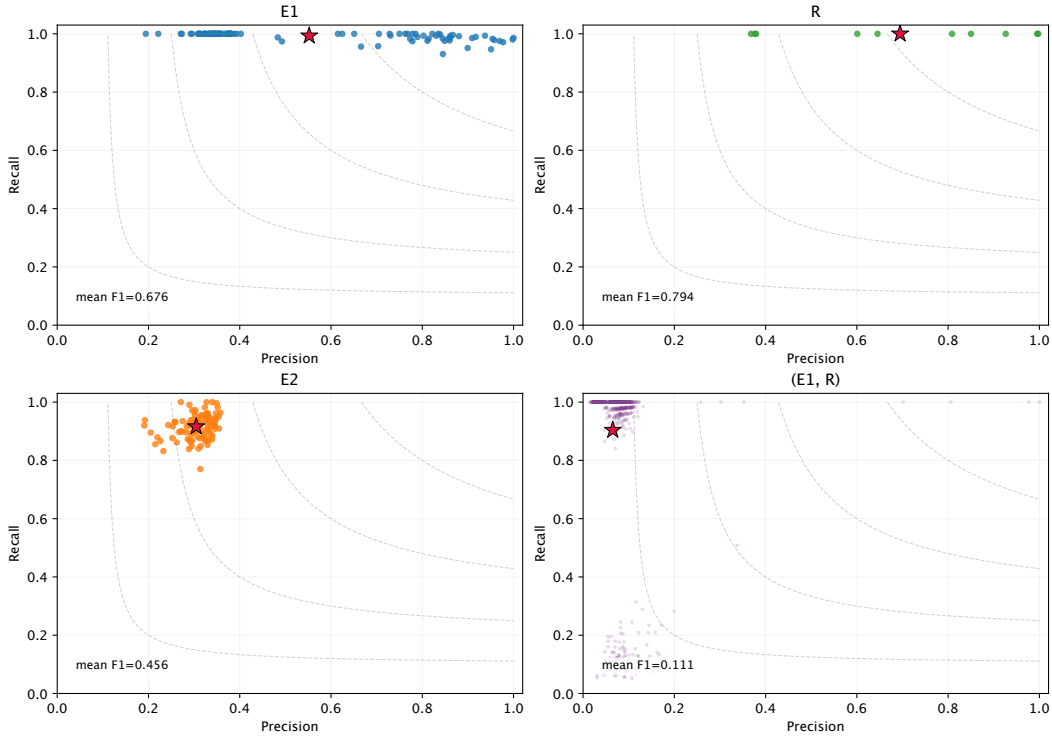


Figure 7. Best-F1 feature recovery for a 2,048-latent SAE trained on the address-carrying Layer 0 head-0 output at [SEP] positions. Panels show results for the marginal labels E_1 , R , E_2 , and the joint address (E_1, R) . Each point is the precision and recall of the highest-F1 SAE feature for one label; the red star marks the per-panel mean.

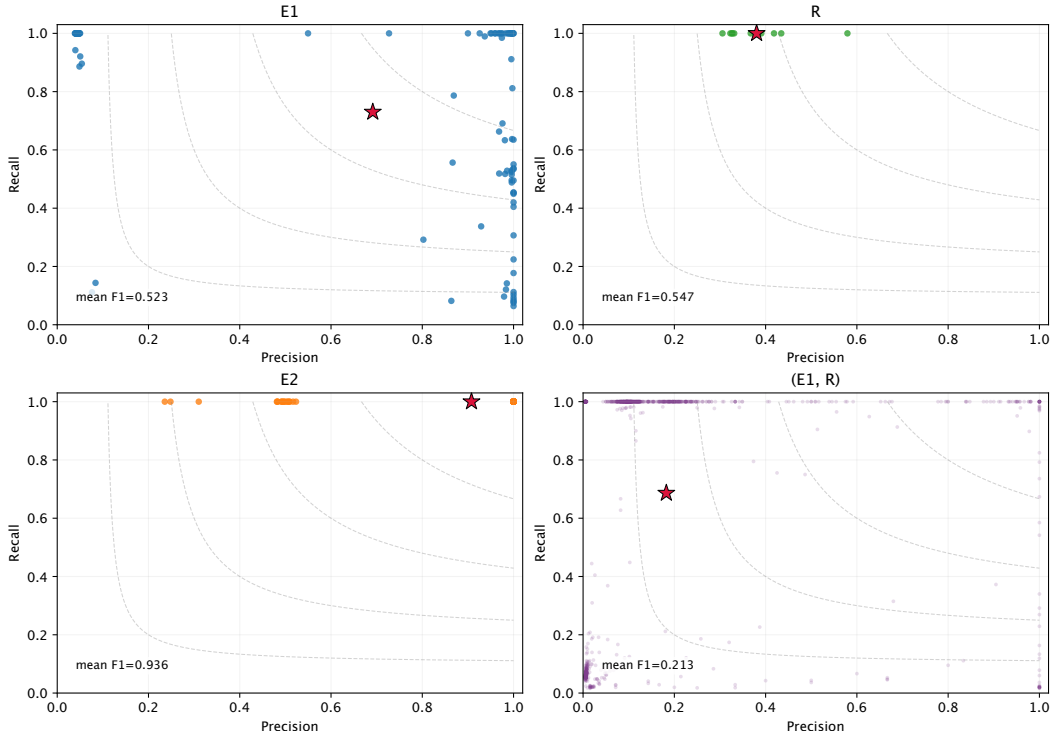


Figure 8. Best-F1 feature recovery for a 2,048-latent SAE trained on the Layer 0 residual stream at [SEP] positions. Panels show results for the marginal labels E_1 , R , E_2 , and the joint address (E_1, R) .